Assignment 7. Fixed-point & Newton-Raphson Marks 20

Posted on 03.09.2025 @ 2:30 pm and due on 03.09.2025 @ 5:40 pm

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1. Write the library functions for Fixed-point and Newton-Raphson methods. **[3+5]**

2. Find the root of the following function in the interval $[-1.5, 1.5]$ to an accuracy of 10^{-6} using Bisection, Regula Falsi and Newton-Raphson methods. Compare their convergence rate. **[4]**

$$f(x) = 3x + \sin(x) - e^x$$

3. Find the root of the following function using Fixed-point method **[3]**

$$f(x) = x^2 - 2x - 3$$

4. Solve the following using either Newton-Raphson or Fixed-point **[5]**

$$f_1(x, y) = 3x^2 - 2xy - 3$$
$$f_2(x, y) = 3y^2 - 4xy$$